Shifty employs **XGBoost Regressor** for pricing as it captures real-world complexities like **peak-time surcharges, seasonal demand, and item-specific handling costs**.

2.3.3 Natural Language Processing (NLP) for Chatbots

The chatbot component enhances user experience through conversational support.

- **Intent Recognition**: Determines user intent (e.g., "track order," "get quote," "reschedule move"). Advanced models like **BERT** or lightweight frameworks like **Rasa NLU** can be used.
- **Entity Extraction**: Identifies entities such as order IDs, dates, or locations from queries.
- **Dialogue Management**: Ensures smooth conversation flow, enabling the bot to answer FAQs, escalate issues, or provide real-time status updates.

Together, these NLP techniques enable the chatbot to provide **24/7 assistance**, reducing dependency on manual customer service.

2.4. Summary

The review of existing systems reveals a lack of AI integration in the current packers and movers ecosystem. While aggregator platforms and individual mover websites provide basic digital interfaces, they do not address critical issues such as personalization, dynamic pricing, and intelligent support.

Machine Learning and NLP offer powerful tools to overcome these limitations. By adopting algorithms such as Random Forest for recommendations, XGBoost for dynamic pricing, and NLP for chatbot support, SHIFTY leverages cutting-edge technology to create a holistic, user-centric platform.

The selected technology stack ensures scalability, flexibility, and performance, positioning SHIFTY as a pioneering solution in the relocation industry.